

PHARMA CO: PREDICTING EMPLOYEE ATTRITION

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Executive Summary

Pharma Co, a pharmaceuticals company, has expressed concerns regarding employee dissatisfaction and staff turnover. Employee attrition can result in significant replacement costs, loss of company knowledge and customer service interruptions (Bhuvaneswari & Kavitha, 2022). In order to address the problem of attrition, Pharma Co has provided a 1000-observation employee dataset, with the binary variable for attrition included.

The purpose of this report is to (1) identify and analyse two predictive models that were built to predict whether an employee will leave Pharma Co, and (2) use the predictors from these models to draw further insights and provide recommendations to address the problem of employee attrition. This report also explains the process by which the two predictive models were selected, and critically discusses the performance of the two predictive models.

As a brief overview, the two models were selected using the process of “forward stepwise selection” (discussed in the “Analysis Methodology” section) with models being evaluated using cross-validated “area under the curve” (AUC) (Appendix B and Appendix C). Final models were evaluated using AUC, accuracy, specificity and sensitivity (outlined in the “Analysis Methodology”, Appendix B and Appendix C).

The two models included in this report are a (1) logistic regression model and (2) decision tree model. Two important predictor variables found to have a relationship with employee attrition include monthly income and overtime – which were common across both models. Additional predictor variables used in the models include marital status, job involvement score, total working years and working environment score. The impact of each predictor has been analysed in the discussion on the models (section “Predictive Model 1 – Logistic Regression” and “Predictive Model 2 – Decision Tree”) and recommendations based on these predictors can be found in the section “Recommendations based on Model Predictors”.

Overall, both models provided useful information to address the problem of employee attrition. Recommendations to consider include implementing alternate forms of compensation such as childcare benefits, implementing tenure-based retention bonuses, cross-training employees, and process/technology changes to reduce unnecessary overtime. These are further elaborated upon in the section “Recommendations based on Model Predictors”.

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Context and Problem Definition

Pharma Co, a pharmaceuticals company, has expressed concerns regarding employee dissatisfaction and staff turnover. Employee attrition can result in significant replacement costs, loss of company knowledge and customer service interruptions (Bhuvaneswari & Kavitha, 2022).

Following on from the previous exploratory data analysis and findings (Gnanarajan, 2024), Pharma Co has provided an additional column within the dataset – “Attrition” – which has two values (“Yes” and “No”). The company requests for the full dataset to be used to build models to predict whether or not an employee will leave the organisation, and for insights to be drawn based on the predictor variables used in the models.

To address Pharma Co’s problem and requirements, this report will include the following:

- The identification of 2 selected predictive models
- Identification and discussion of predictors used in the 2 selected models
- Recommendations based on the predictors used in the models

Additionally, this report will outline the process used to select the models, and discuss the performance of these models.

Analysis Methodology

The full dataset provided by Pharma Co was analysed using the R-programming language. The full R-code can be found in **Appendix A**.

Since the target “Attrition” variable is a binary variable (with exactly 2 values – “Yes” and “No”), the 2 types of models used for prediction of attrition include:

1. Logistic regression – a model that fits an S-shaped curve to its training data
2. Decision tree – a model that partitions its training data into buckets

These models were used to predict the *probability of attrition*.

Evaluation of Model Performance

The performance of various models was evaluated using the measures of accuracy, sensitivity, specificity, “Area Under the Curve” (AUC) and cross-validated AUC. The technical details of these measures can be found in **Appendix B** (and **Appendix C** for cross-validation), but as a brief overview:

- **Accuracy** is the percentage of cases overall which were predicted correctly by the model (including attrition and retention). This can range from 0% to 100%.
- **Sensitivity** is the proportion of true cases of employee attrition which were predicted correctly by the model (ie. accuracy on positive cases). This can range from 0 (worst) to 1 (best).
- **Specificity** is the proportion of cases of employee retention (ie. “Attrition” = 0) which were predicted correctly by the model (ie. accuracy on negative cases). This can range from 0 (worst) to 1 (best).
- **AUC** is the “**area under the curve**” that results when “sensitivity” and “specificity” are plotted against each other, across varying decision probability thresholds. This curve gives a picture of how much the “sensitivity” and “specificity” of the model can be maximised. This can range from 0 (worst) to 1 (best).

- **Cross-validated AUC (10-fold)** is the average AUC across 10 folds of test data, after the process of 10-fold cross-validation has been carried out. This can range from 0 (worst) to 1 (best). Technical details of cross-validation can be found in **Appendix C**.

Forward Stepwise Selection

The final selected logistic regression model and decision tree model were found for this report using a method called “Forward Stepwise Selection” (UNSW, 2024) – illustrated in **Figure 1** below. This is a process by which the model starts with 0 predictor variables. One by one, the “best” of the available predictor variables is added to the model, as long as the performance of the model keeps improving. In this scenario, the performance measure was the *10-fold cross-validated AUC*; it was used to evaluate which predictor variable was “best” to add to the model (ie. provided the greatest increase in cross-validated AUC), and when to stop adding additional predictor variables.

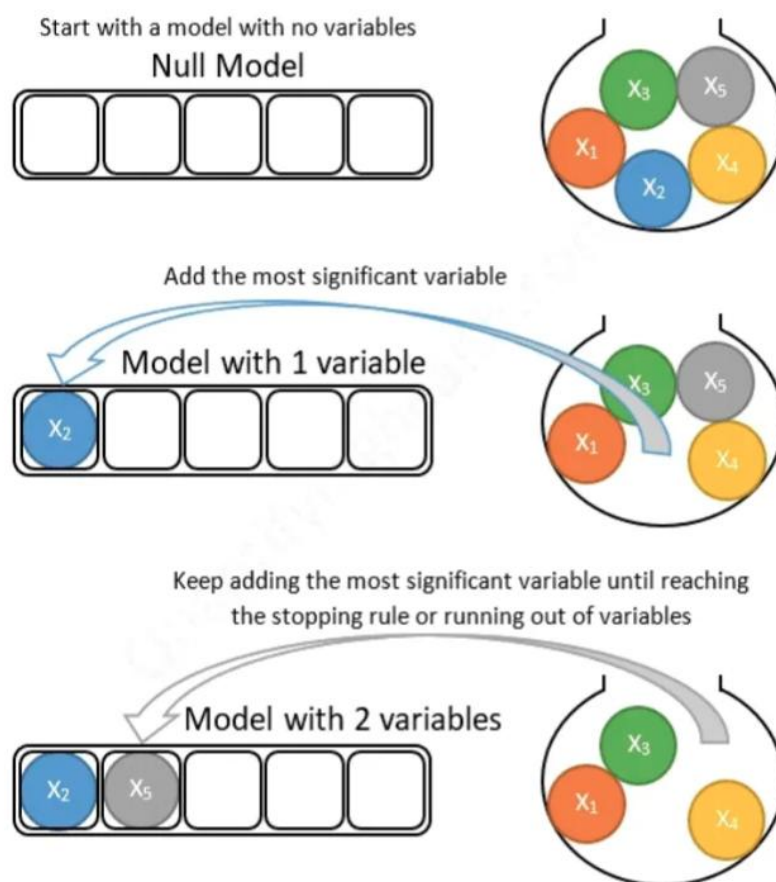
In our scenario, the “stopping rule” shown in **Figure 1** below was when the cross-validated AUC could not be increased by more than 0.01 (with the addition of an extra variable). In other words, additional predictor variables would be added to the model as long as:

Cross-validated AUC with added predictor > (Cross-validated AUC with current predictors + 0.01)

The 0.01 adjustment was used as a way to keep the models simpler with fewer predictors, so that the insights of this report could be focussed on the most important predictors.

Figure 1: Forward Stepwise Selection (Frenzel, 2024)

Forward stepwise selection example with 5 variables:



Analysis of Attrition Models, Predictors and Performance

Predictive Model 1 – Logistic Regression

The first model is a logistic regression model (ie. a model fits an S-shaped curve to its training data), where the probability of attrition is predicted by the equation:

$$Probability(Attrition) = \frac{1}{1 + e^{-(B_0 + B_1x_1 + B_2x_2 + B_3x_3 + B_4x_4 + B_5x_5)}}$$

The variables (x_n) and coefficients (B_n) are given as per **Table 1** below. The “Likelihood of Genuine Relationship” is based on the “significance codes” that are provided with the model outputs in R. The full technical model summary from R can be obtained by running the code in **Appendix A**.

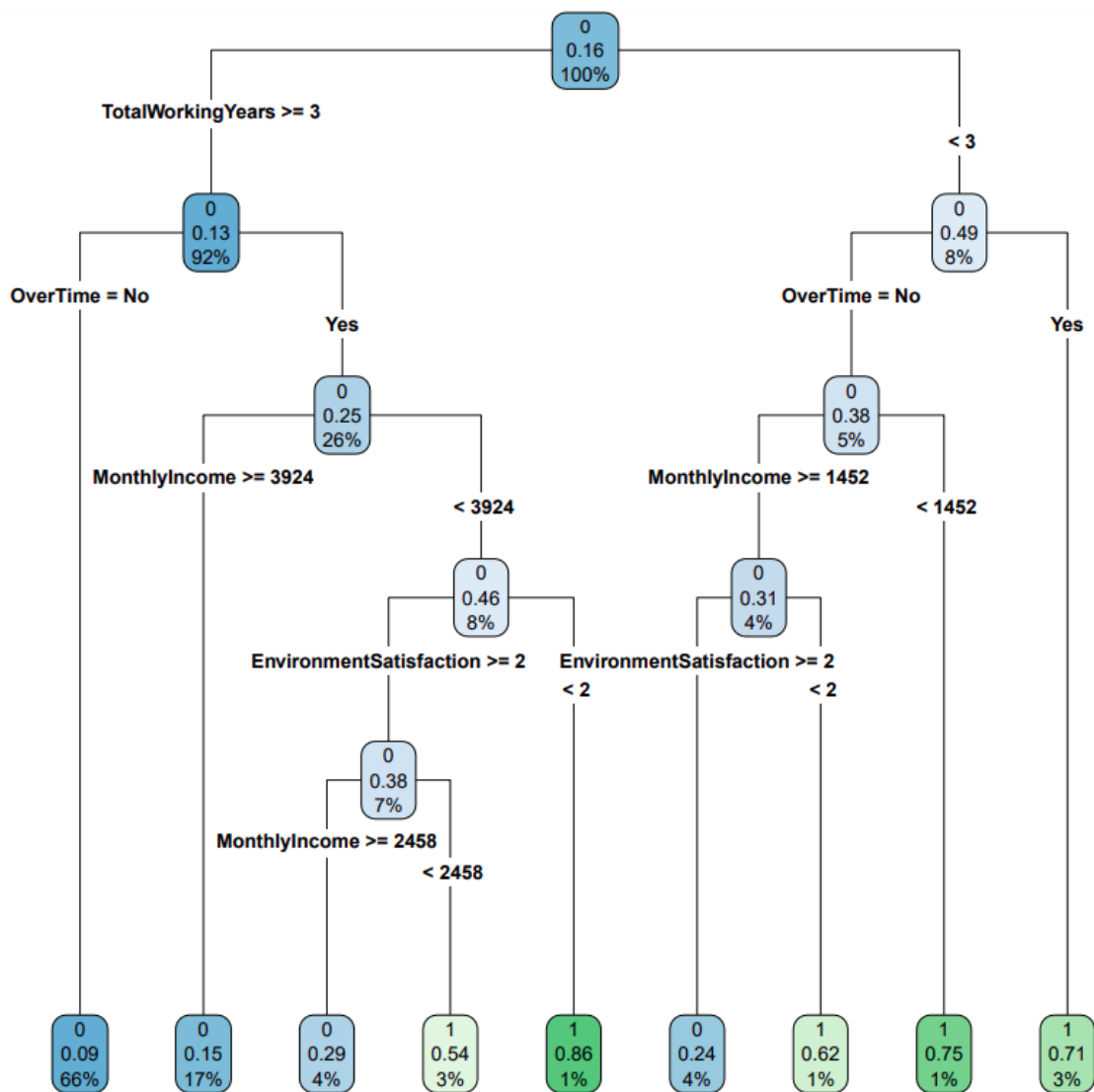
Table 1: Logistic Regression Model – Variables and Coefficients

x_n (variable)	B_n (coefficient)	Likelihood of Genuine Relationship (with attrition)	Explanation
(Intercept)	-6.32E-01	N/A	
MonthlyIncome	-1.44E-04	High	An employee with a lower income is more likely to leave than an employee with a higher income, considering all other factors equal.
OverTimeYes	1.42E+00	High	An employee who works overtime is more likely to leave than an employee who doesn't work overtime, considering all other factors equal.
MaritalStatusMarried	5.05E-01	Moderate	An employee who is married is more likely to leave than an employee who is unmarried, considering all other factors equal.
MaritalStatusSingle	1.37E+00	High	An employee who is single is more likely to leave than one who isn't, considering all other factors equal.
JobInvolvement	-5.66E-01	High	An employee with a lower job involvement score is more likely to leave than an employee with a higher score, considering all other factors equal.

Predictive Model 2 – Decision Tree

The second model is a decision tree model, i.e. a model that partitions its training data into buckets based on characteristics. The decision tree model is shown in the diagram in **Figure 2** below:

Figure 2: Decision Tree Model



The bright green buckets indicates those most likely to attrite. The predictors used in the model are:

- **TotalWorkingYears** (variable importance 34)
 - It is clear that the vast majority of employees who are most likely to stay are those who have worked a total of 3 or more years, as seen by the 3 leftmost leaves of the tree containing over 85% of total employees.
 - The variable “TotalWorkingYears” provides an important split that enables further splitting of buckets through monthly income. Intuitively, a person’s income increases with their overall work experience.
- **MonthlyIncome** (variable importance 33)
 - Employees with lower incomes are more likely to leave the organisation than those who have higher incomes (considering other factors equal).
- **Overtime** (variable importance 22)
 - Employees doing overtime are more likely to leave than those not doing overtime (considering other factors equal).
- **EnvironmentSatisfaction** (variable importance 11)
 - Employees with lower scores for environment satisfaction are more likely to leave (considering other factors equal).

Analysis of Performance of Model 1 and Model 2

The performance of the two models can be quantitatively evaluated and compared using the cross-validated AUC, AUC, sensitivity, specificity, and accuracy – all of which were outlined in the “Analysis Methodology” section. The sensitivity, specificity and accuracy are based on the finalised models outlined previously (which were trained on the full dataset), and are calculated at the optimal decision probability (0.196) as per the ROC curves shown in **Figure 3** and **Figure 4** below. These measures are shown in **Table 2** below.

Table 2: Performance Measures for Models 1 and 2

	Model 1 (Logistic Regression)	Model 2 (Decision Tree)	Comparison
10-fold Cross-validated AUC	0.76	0.72	Model 1 is likely to be better for prediction with new data
AUC (finalised model)	0.7713	0.7309	Model 1 is better for maximising sensitivity and specificity
Sensitivity	0.660	0.488	Model 1 has better accuracy on cases where people attrite
Specificity	0.757	0.896	Model 2 has better accuracy on cases where people don't attrite
Accuracy	74%	83%	Model 2 has better accuracy overall, mainly due to its high accuracy on cases where people don't attrite
ROC Curve	Figure 3	Figure 4	

The focus of this report is the *predictor variables* rather than *predictions*; both models provide useful predictors for drawing insights. For the purpose of making predictions:

- Model 1 is likely to be better at predicting whether a person will attrite due to the higher cross-validated AUC, and the higher sensitivity.
- If management wanted to identify every single employee who will leave in the following year using a model, it is unlikely that either of these two models will be able to predict this perfectly (on new data), as they currently have sensitivities of 0.660 and 0.488 on the existing data.

Figure 3: ROC Curve for Logistic Regression Model

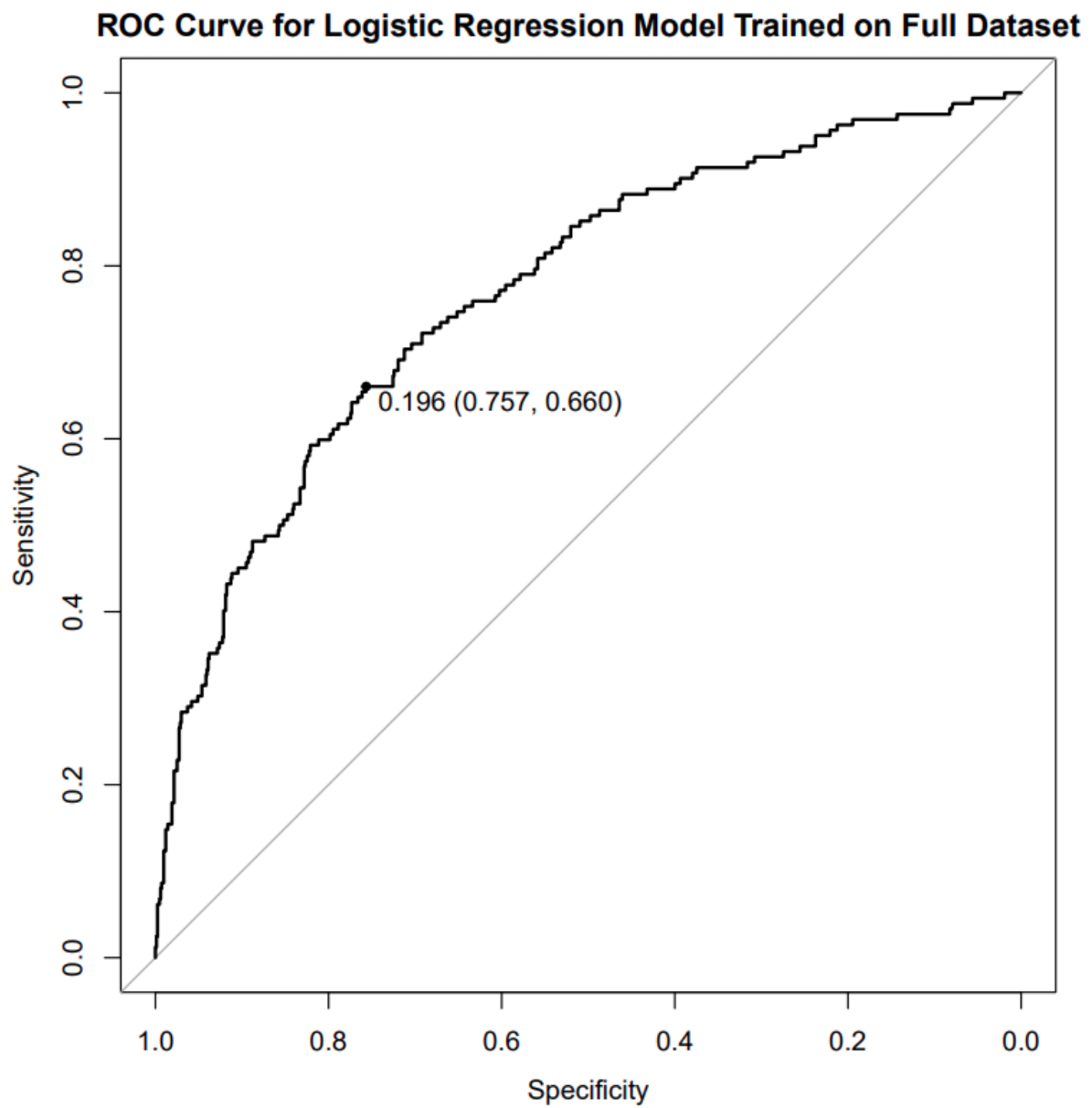
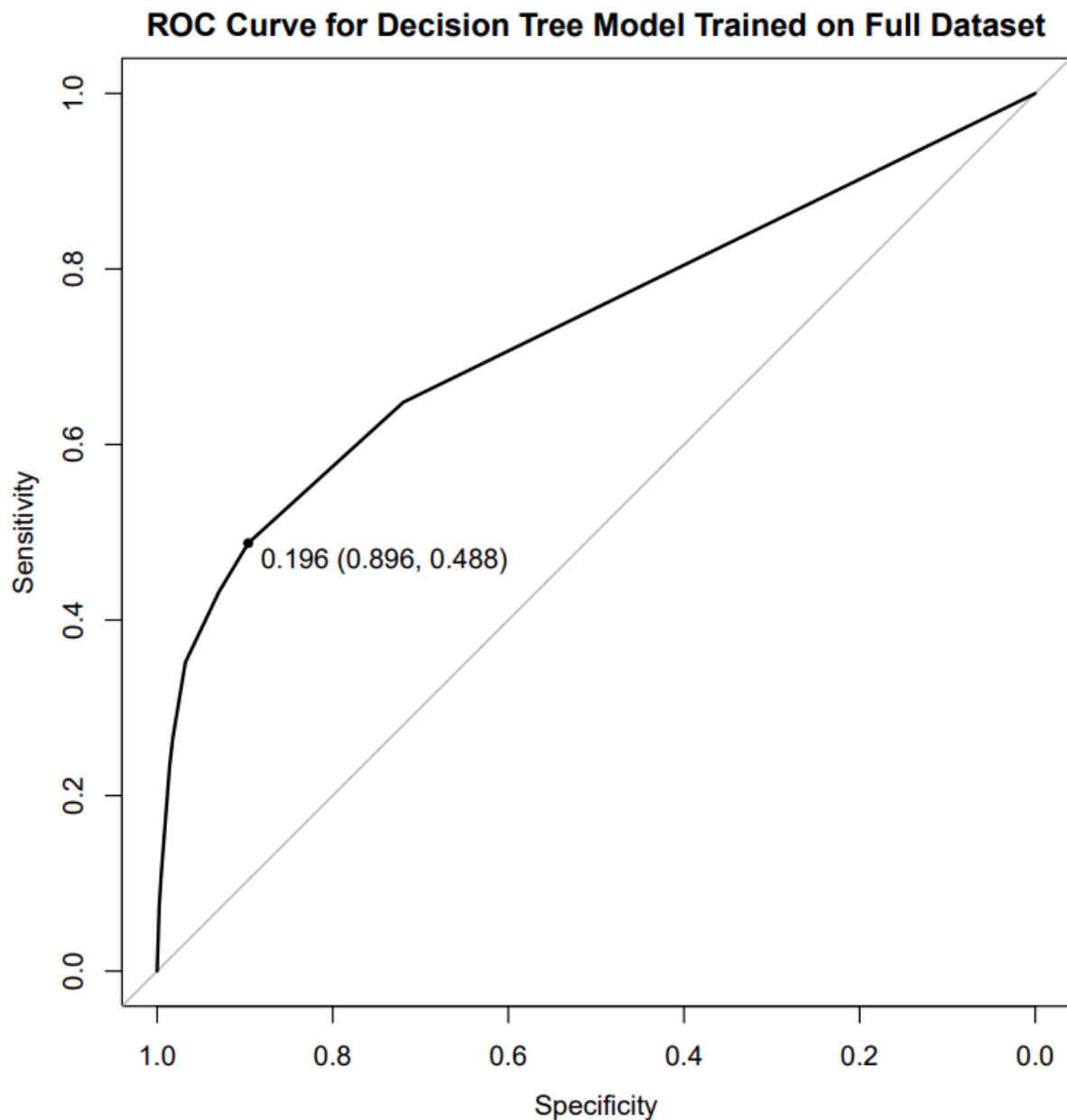


Figure 4: ROC Curve for Decision Tree Model



Limitations of Models

Relationship rather than causation: An important point to note with both the above models is that they cannot conclusively tell us that any particular variables are *causing* attrition. These models tell us that there is a relationship between the selected predictor variables and the target variable (“Attrition”), but this relationship could be due to other underlying factors. For example, employees with lower salaries may be more likely to attrite due to being treated poorly by management (rather than due to the salary itself).

Trained based on provided dataset: The models above have been trained based on the dataset provided with 1000 observations. If it turned out that Pharma Co actually had 10,000 employees, then this would mean the models and predictors were selected based on only 10% of employees – which could have biases. This could mean that the model would not necessarily be as effective in predicting attrition for the remaining 9,000 employees. In general, we cannot definitively know how the model would perform with

new data, outside the 1000-observation dataset provided. If the 1000-person dataset provided is an unbiased and holistic representation of the organisation, then it is likely the model will perform well with new data, but it is not a certainty.

Common Predictors across Both Models

Two important variables that both models have in common are “Monthly Income” and “Overtime”. This supports the idea that there is genuinely a relationship between “Attrition” and these variables. Visually these relationships can be seen by plotting the percentage of people leaving/staying, grouped by these predictor variables. This is shown in **Figure 5** and **Figure 6** below.

Figure 5: Employee Attrition by Monthly Income Bucket

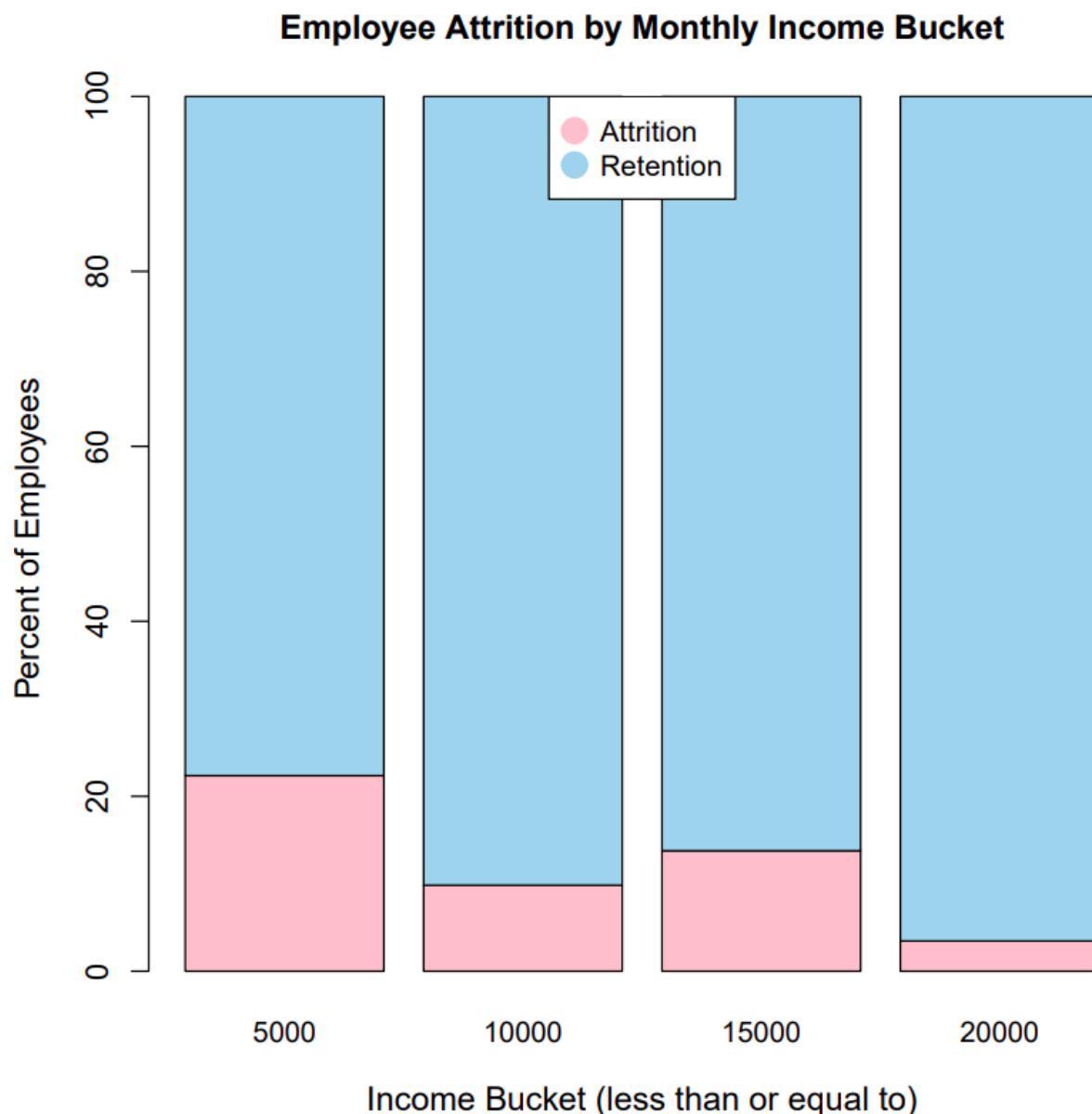
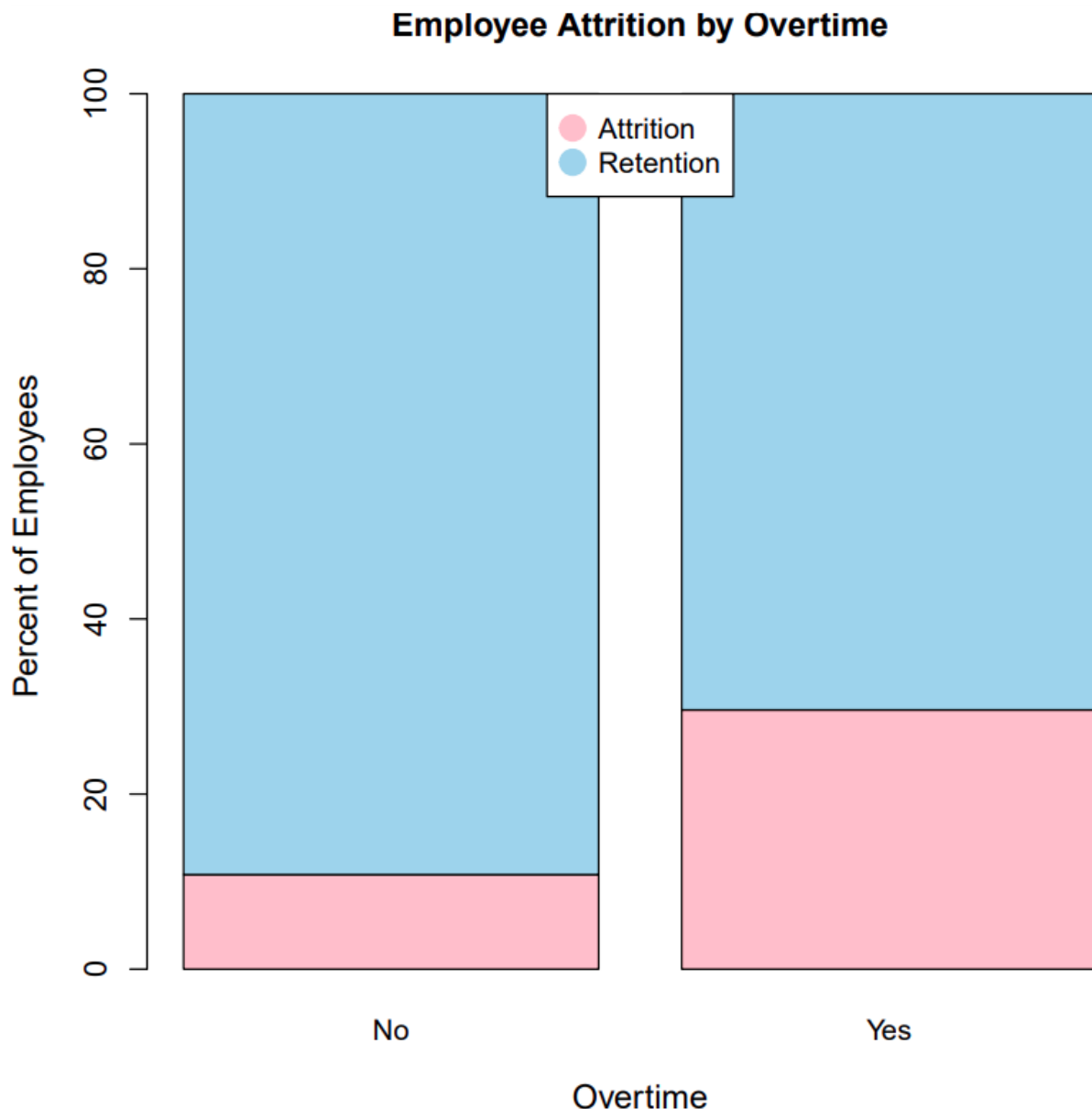


Figure 6: Employee Attrition by Overtime



Recommendations based on Model Predictors

For each of the predictors mentioned in the models above, as discussed in the section “Limitations of Models”, we cannot be certain that they are *causal*. Ideally, to address attrition within the organisation, management should:

1. Investigate whether predictors are likely to be causal, or if any underlying reasons can be uncovered that explain the relationships in the data. This could be through carrying out additional surveys or interviews (Verve Partners, 2022) with the higher-risk groups (as summarised in **Table 3** below).
2. If causality is likely, recommended actions (based on multiple published sources) are provided in **Table 3** below. Note that a cost-benefit analysis would need to be carried out for any recommendations which require funding.

Table 3: Recommendations based on Predictor Variables

Predictor Variable	Risk group (more likely to attrite)	Potential Actions to Consider <i>(if predictor variable is proven likely to be causal)</i>
Monthly Income (top priority)	Employees with lower incomes	- Investigate if any lower-income employees are genuinely underpaid in terms of their productivity/outcomes/expertise, and consider reviewing the compensation and rewards program accordingly (deBara, n.d.). - Consider alternate employee benefits such as childcare or nursing services. There is evidence that companies can retain 100% of mothers with on-site childcare and other family benefits. For Globex, this could mean retaining 1.5-2% of their workforce who would have otherwise left (De Smet, et al., 2021). - Implement a retention bonus or other tenure-based incentives (deBara, n.d.).
Overtime (top priority)	Employees who work overtime	- Investigate main factors driving the need for overtime work (eg. SLAs, targets or policies) and review accordingly (Roman, 2023). - Implement process changes or automated solutions to reduce unnecessary overtime (Roman, 2023). - Cross-train employees to reduce single-person dependencies (Celayix, 2024) (Roman, 2023) - Reduced overtime could result in a 33% increase in employee productivity for affected individuals (Celayix, 2024) and improve retention outcomes for up to 8.5% of Globex' workforce.
Marital Status (lower priority)	Employees who are single	Investigate potential unconscious bias in the workplace where single workers are disadvantaged (Evbuomwan , 2021), and implement training programs accordingly. This could improve retention outcomes for up to 8.4% of Globex' employees.
Total Working Years (lower priority)	Employees who have worked fewer (<3) years	- Interview employees on what factors would help them stay at Globex long-term. - Revise onboarding experience as studies show this can improve new hire retention by 82% (Victoria University, 2024)
Job Involvement & Environment Satisfaction (lower priority)	Employees with lower scores for job engagement / environment satisfaction	<i>(These variables may be influenced by other variables that have already been discussed.)</i> - Review employee objectives and ensure they are connected to organisation's mission statement. (Stein, et al., 2021) - Ensure in-person meetings take place to foster a sense of community (Victoria University, 2024) (De Smet, et al., 2021) - Implement workplace employee feedback system (Victoria University, 2024)

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Appendices

Appendix A – Method of Analysis (Code)

```
# Predicting employee attrition for Pharma Co
# Anne Gnanarajan
suppressMessages(library(pROC))
library(rpart)
library(rpart.plot)

# Reading employee data

employees <- read.csv(
  file = '/course/data/employees2.csv',
  stringsAsFactors = TRUE
)

head(employees)
str(employees)

# Changing attrition to 0/1 (for modelling purposes) rather than
"Yes"/"No"
employees$Attrition <- ifelse(employees$Attrition == "Yes", 1, 0)

str(employees)

# Create the ten folds
#####
set.seed(57)
employees$Fold <- sample(rep(1:10, length.out=nrow(employees)))

# Define a function to do the cross-validation
# LOGISTIC REGRESSION MODEL
#####
####
cv_score <- function(predictors) {
  scores <- c()
  for (fold in 1:10) {
    training = employees[employees$Fold != fold,]
    validation = employees[employees$Fold == fold,]
    model <- glm(
      # Build the formula from the predictors
      formula = paste("Attrition ~ ", paste(predictors, collapse = " +
"), sep = ""),
      data = training,
      family = "binomial"
    )
```



```

validation$Probability <- predict(
  object = model,
  newdata = validation,
  type = "response"
)
auc <- roc(validation$Attrition ~ validation$Probability, quiet =
TRUE)$auc
scores <- c(scores, auc)
#message(paste("Predictors ",predictors, "Fold ", fold, sep = ""))
}
return(mean(scores))
}

# Define a function to do the stepping forward
#####
####
step_forward <- function(current_predictors, possible_predictors) {

  # If there are no possible predictors, return the current_predictors
  if (length(possible_predictors) == 0) return(current_predictors)

  # Get the cross-validation score of the current predictors
  if (length(current_predictors) > 0) current_score <-
cv_score(current_predictors)
  else current_score = 0

  # Show the current predictors and their cross-validation score
  message("Current predictors: ", paste(current_predictors, collapse="
+ "))
  message("Current score: ", current_score)

  # Create a vector in which to put the new scores
  new_scores <- c()

  # Try adding each of the possible predictors
  for (predictor_to_try in possible_predictors) {
    # Add the predictor to the current predictors
    new_predictors <- c(current_predictors, predictor_to_try)
    # Find the cross-validation score of the new predictors
    new_score <- cv_score(new_predictors)
    # Show the score
    message("With ", predictor_to_try, ": ", new_score)
    # Add it to the vector of new scores
    new_scores[predictor_to_try] <- new_score
  }

  # If the best new score is better than the current score
  if (max(new_scores) > (current_score + 0.01)) {

```

```

# Find the predictor to add
predictor_to_add <- names(new_scores)[which.max(new_scores)]
# Output an explanatory message
message("Adding ", predictor_to_add, "\n")
# Add it to the current predictors
current_predictors <- c(current_predictors, predictor_to_add)
# Remove it from the possible predictors
possible_predictors <- possible_predictors[possible_predictors !=
predictor_to_add]
# Step forward again
return(step_forward(current_predictors, possible_predictors))
}

# Otherwise
else {
  message("Nothing to add\n")
  return(current_predictors)
}
}

# Define the starting predictors and the possible predictors
#####
###
starting_predictors <- c()
possible_predictors <- c(
  'Age',
  'BusinessTravel', #### 5
  'Department',
  'DistanceFromHome',
  'Education',
  'EnvironmentSatisfaction',
  'JobInvolvement', #### 4
  'JobRole',
  'MaritalStatus', #### 3
  'MonthlyIncome', #### 1
  'NumCompaniesWorked',
  'PerformanceRating',
  'RelationshipSatisfaction',
  'StockOptionLevel',
  'TotalWorkingYears',
  'TrainingTimesLastYear',
  'WorkLifeBalance',
  'YearsAtCompany',
  'YearsInCurrentRole',
  'YearsSinceLastPromotion',
  'YearsWithCurrManager',
  'Gender',
  'OverTime' #### 2

```

```

    )
print(possible_predictors)

# Use step_forward to find the best predictors
best_predictors <- step_forward(starting_predictors,
possible_predictors)

# Show the results
message("Best predictors: ", paste(best_predictors, collapse = " + "))
message("Cross-validation score: ", cv_score(best_predictors))

##### Training the model with the final selection of variables,
across the full dataset
#####
model <- glm(
  formula = Attrition ~ MonthlyIncome + OverTime + MaritalStatus +
JobInvolvement, #best predictors
  data = employees,
  family = "binomial"
)
model
summary(model)

#final predictions
employees$Probability <- predict(
  object = model,
  newdata = employees,
  type = "response"
)
head(employees)

results <- roc(
  employees$Attrition ~ employees$Probability,
  plot = TRUE,
  print.thres = "best",
  main = "ROC Curve for Logistic Regression Model Trained on Full
Dataset")
results

#Create predictions based on best threshold
employees$Prediction <- ifelse(employees$Probability>= 0.196, 1, 0) #
best threshold
#Create confusion matrix
conf_matrix <- table(employees$Prediction, employees$Attrition)
conf_matrix
# Calculate the accuracy
accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)

```

```

# Show the accuracy
message("Accuracy: ", accuracy)

### Decision Tree Model
#####

# Define a function to do the cross-validation
#####
###
cv_score <- function(predictors) {
  scores <- c()
  for (fold in 1:10) {
    training = employees[employees$Fold != fold,]
    validation = employees[employees$Fold == fold,]
    model <- rpart(
      # Build the formula from the predictors
      formula = paste("Attrition ~ ", paste(predictors, collapse = " +
"), sep = " "),
      data = training,
      method = "class",
      maxdepth = 5
    )
    validation$Probability <- predict(
      object = model,
      newdata = validation,
      type = "prob"
    )
    auc <- roc(validation$Attrition ~ validation$Probability[,2], quiet
= TRUE)$auc
    scores <- c(scores, auc)
    #message(paste("Predictors ",predictors, "Fold ", fold, sep = ""))
  }
  return(mean(scores))
}

# Use step_forward to find the best predictors
best_predictors <- step_forward(starting_predictors,
possible_predictors)

# Show the results
message("Best predictors: ", paste(best_predictors, collapse = " + "))
message("Cross-validation score: ", cv_score(best_predictors))

##### Training the model with the final selection of variables,
across the full dataset
#####
model <- rpart(

```

```

    formula = Attrition ~ MonthlyIncome + OverTime + TotalWorkingYears +
EnvironmentSatisfaction, # best predictors
    data = employees,
    method = "class",
    maxdepth = 5
)
model
summary(model)

# final predictions
employees$Probability <- predict(
  object = model,
  newdata = employees,
  type = "prob"
)
head(employees)

results <- roc(
  employees$Attrition ~ employees$Probability[,2],
  plot = TRUE,
  print.thres = "best",
  main = "ROC Curve for Decision Tree Model Trained on Full Dataset")
results

#Plotting decision tree
rpart.plot(
  model,
  type = 4
)

#Create predictions based on best threshold
employees$Prediction <- ifelse(employees$Probability[,2]>= 0.196, 1, 0)
# best threshold
#Create confusion matrix
conf_matrix <- table(employees$Prediction, employees$Attrition)
conf_matrix
# Calculate the accuracy
accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)
# Show the accuracy
message("Accuracy: ", accuracy)

##### Plotting #####
#####

## Plotting Monthly Income and Attrition
# Creating employee income brackets
employees$IncomeBracket[employees$MonthlyIncome<=5000] <- 5000

```

```

employees$IncomeBracket[employees$MonthlyIncome>5000 &
employees$MonthlyIncome<=10000] <- 10000
employees$IncomeBracket[employees$MonthlyIncome>10000 &
employees$MonthlyIncome<=15000]<- 15000
employees$IncomeBracket[employees$MonthlyIncome>15000 &
employees$MonthlyIncome<=20000]<- 20000
employees$IncomeBracket[employees$MonthlyIncome>20000 &
employees$MonthlyIncome<=25000]<- 25000

head(employees)
# Summarising employees by Income and Attrition
income_attrition <- aggregate(
  JobRole ~ IncomeBracket + Attrition,
  data = employees,
  FUN = length
)
income_attrition

#combining table for males with table for females for plotting
income_attrition1 <- merge(
  x = income_attrition[income_attrition$Attrition==0,],
  y = income_attrition[income_attrition$Attrition==1,],
  by = "IncomeBracket",
  all = TRUE)
income_attrition1

# Calculating percentages
income_attrition1$Retention <-
(income_attrition1$JobRole.x/(income_attrition1$JobRole.x+income_attrit
ion1$JobRole.y))*100
income_attrition1$Attrition <-
(income_attrition1$JobRole.y/(income_attrition1$JobRole.x+income_attrit
ion1$JobRole.y))*100

income_attrition1

# creating bar plot
barplot(
  cbind(Attrition, Retention) ~ IncomeBracket,
  data = income_attrition1,
  col = c('pink', 'lightskyblue2'),
  ylab = "Percent of Employees",
  xlab = "Income Bucket (less than or equal to)",
  #las = 3,
  main = "Employee Attrition by Monthly Income Bucket",
  cex.main = 1.2, cex.lab = 1.2, cex.axis = 1.1
)

```

```

# Adding legend for colours
legend(
  # Put the legend in the top middle
  "top",
  legend = c("Attrition", "Retention"),
  col = c("pink", "lightskyblue2"),
  pch = 19,
  pt.cex = 2,
  bg = "white"
)

# Summarising employees by Overtime and Attrition
overtime_attrition <- aggregate(
  JobRole ~ OverTime + Attrition,
  data = employees,
  FUN = length
)
overtime_attrition

#combining table for males with table for females for plotting
overtime_attrition1 <- merge(
  x = overtime_attrition[overtime_attrition$Attrition==0,],
  y = overtime_attrition[overtime_attrition$Attrition==1,],
  by = "OverTime",
  all = TRUE)
overtime_attrition1

# Calculating percentages
overtime_attrition1$Retention <-
(overtime_attrition1$JobRole.x/(overtime_attrition1$JobRole.x+overtime_
attrition1$JobRole.y))*100
overtime_attrition1$Attrition <-
(overtime_attrition1$JobRole.y/(overtime_attrition1$JobRole.x+overtime_
attrition1$JobRole.y))*100

overtime_attrition1

# creating bar plot
barplot(
  cbind(Attrition, Retention) ~ OverTime,
  data = overtime_attrition1,
  col = c('pink', 'lightskyblue2'),
  ylab = "Percent of Employees",
  xlab = "Overtime",
  #las = 3,
  main = "Employee Attrition by Overtime",
  cex.main = 1.2, cex.lab = 1.2, cex.axis = 1.1
)

```

```

legend(
  # Put the legend in the top middle
  "top",
  legend = c("Attrition", "Retention"),
  col = c("pink", "lightskyblue2"),
  pch = 19,
  pt.cex = 2,
  bg = "white"
)

```

Appendix B – Accuracy, Sensitivity, Specificity and AUC

As per the UNSW Course notes (UNSW, 2024):

Accuracy can be calculated as the “proportion of predictions that are correct”. It is defined as follow:

$$\text{Accuracy} = \frac{\text{Number of true predictions}}{\text{Total number of predictions}}$$

Sensitivity is the “model’s accuracy on positive cases”. It can be calculated as:

$$\text{Sensitivity} = \text{Accuracy on positive cases} = \frac{TP}{TP + FN}$$

Specificity is the “model’s accuracy on negative cases”. It can be calculated as:

$$\text{Specificity} = \text{Accuracy on negative cases} = \frac{TN}{TN + FP}$$

An **ROC curve** a graph showing “Sensitivity” plotted against “Specificity” at all classification thresholds (**Figure 7** below). Alternately, the “True Positive Rate” (TPR) can be plotted against the “False Positive Rate” (FPR), where the FPR is: $(1 - \text{Specificity})$, with the X-axis flipped. Either way, the shape of the resulting curve is the same and can be interpreted as per **Figure 8** below.

The **Area Under the Curve (AUC)** is the area under the ROC curve. The “better” the ROC curve, the higher the AUC (with the maximum being 1).

Figure 7: ROC Curve (modified from original source) (Google, 2022)

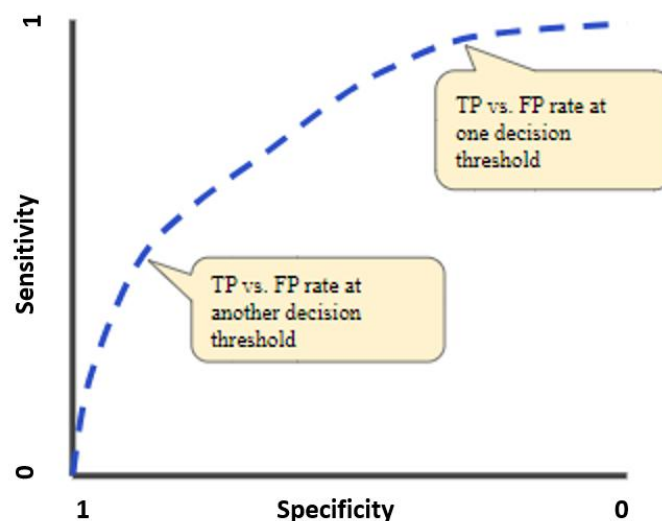
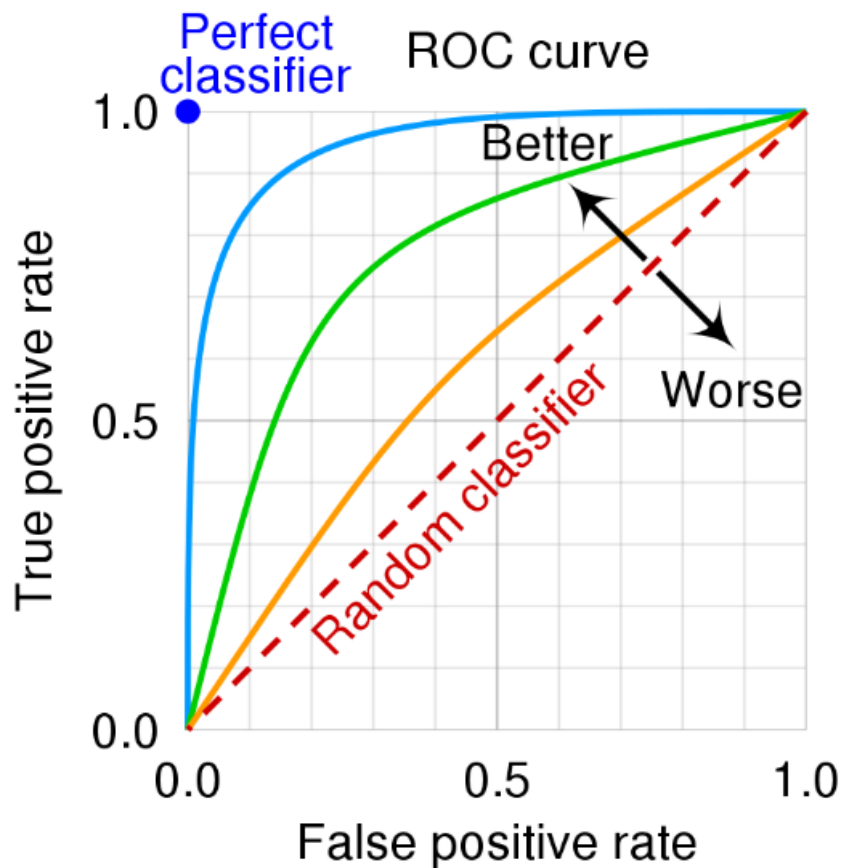


Figure 8: ROC Curves Interpretation (Thoma, 2024)



Appendix C – Ten-fold Cross-validation

In addition to the above measures, the models were evaluated based on “10-fold cross validation” – illustrated in **Figure 9** Below. This a process where the full dataset is divided into 10 folds. Each fold is iterated through as being the “test” set while the remaining data is used to train the model. Ultimately the 10-fold cross validated AUC is calculated as a measure of how well the model can make predictions on new data.

Figure 9: 10-fold Cross Validation (Nellihela, 2022)

